

# M5: Variable Acceleration

## 1. The Calculus Connection

When acceleration is not constant, the constant acceleration (suvat) equations do not apply. Instead, we use calculus to relate displacement, velocity, and acceleration.

**Velocity from displacement:**

$$v = \frac{ds}{dt}$$

**Acceleration from velocity:**

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

**Alternative acceleration form:**

$$a = v \frac{dv}{ds} \text{ (useful when } a \text{ depends on } s \text{)}$$

Integration reverses differentiation:

$$s = \int v \, dt + C \quad v = \int a \, dt + C$$

Always use an **initial condition** (e.g.,  $s = 0$  when  $t = 0$ ) to find the constant of integration  $C$ , or use definite integrals with limits.

## 2. Direction of Motion

The sign of velocity indicates direction (e.g., positive = rightward/upward):

- **Particle at rest:**  $v = 0$
- **Particle reverses direction:**  $v$  changes sign; the maximum displacement from the starting point occurs when  $v = 0$

- **Maximum speed:** occurs when  $\frac{dv}{dt} = a = 0$  (but check it is a maximum using the second derivative or context)

### 3. Finding Displacement Using Integration

To find displacement between times  $t_1$  and  $t_2$ :

$$\text{Displacement} = \int_{t_1}^{t_2} v \, dt$$

To find total **distance** travelled (not displacement), split the integral at times when  $v = 0$  and take the absolute value of each part:

$$\text{Distance} = \int_{t_1}^{t_2} |v| \, dt$$

**Exam Trap:** Displacement and distance are different. If the particle reverses, the integrals can partially cancel — always check for sign changes in  $v$  and compute distance as the sum of absolute values of each segment.

### 4. Worked Example — Full Solution

A particle moves so that its velocity at time  $t$  seconds is given by:

$$v = 3t^2 - 12t + 9 \quad (t \geq 0)$$

The particle starts at the origin ( $s = 0$  when  $t = 0$ ).

**(a) When is the particle at rest?**

$$\text{Set } v = 0: 3t^2 - 12t + 9 = 0 \Rightarrow t^2 - 4t + 3 = 0 \Rightarrow (t - 1)(t - 3) = 0$$

At rest when  $t = 1$  s and  $t = 3$  s.

**(b) Find the acceleration at  $t = 2$ .**

$$a = \frac{dv}{dt} = 6t - 12$$

At  $t = 2$ :  $a = 12 - 12 = 0 \text{ m/s}^2$  (this is when velocity is minimum, a turning point).

**(c) Find the displacement at  $t = 3$ .**

$$s = \int_0^3 (3t^2 - 12t + 9) dt = [t^3 - 6t^2 + 9t]_0^3 = (27 - 54 + 27) - 0 = 0 \text{ m}$$

(Particle returns to origin at  $t = 3$ .)

**(d) Find the total distance travelled from  $t = 0$  to  $t = 3$ .**

Check signs of  $v$ : at  $t = 0$ ,  $v = 9 > 0$ ; at  $t = 2$ ,  $v = 12 - 24 + 9 = -3 < 0$ .

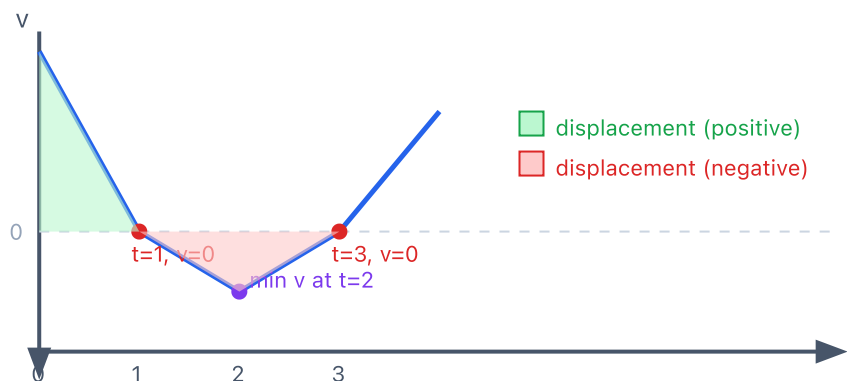
So  $v > 0$  on  $[0, 1]$  and  $v < 0$  on  $[1, 3]$ .

$$s_1 = \int_0^1 v dt = [t^3 - 6t^2 + 9t]_0^1 = 1 - 6 + 9 = 4 \text{ m}$$

$$s_2 = \int_1^3 v dt = (0) - 4 = -4 \text{ m (magnitude 4 m)}$$

Total distance =  $4 + 4 = 8 \text{ m}$

## 5. v-t Graph Interpretation



v-t graph for  $v = 3t^2 - 12t + 9$ : green area = positive displacement, red area = negative displacement

## 6. The Alternative Form: $a = v \frac{dv}{ds}$ [Further Maths — beyond the Edexcel A-level Maths spec]

This form is used when acceleration is given as a function of displacement  $s$ , or when you need to find velocity as a function of position.

**Example:** A particle has acceleration  $a = 2s + 1 \text{ m/s}^2$ . Given  $v = 3$  when  $s = 0$ , find  $v$  when  $s = 2$ .

Use  $v \frac{dv}{ds} = 2s + 1$

Integrate:  $\frac{v^2}{2} = s^2 + s + C$

At  $s = 0, v = 3: C = \frac{9}{2}$

At  $s = 2: \frac{v^2}{2} = 4 + 2 + 4.5 = 10.5 \Rightarrow v^2 = 21 \Rightarrow v = \sqrt{21} \approx 4.58 \text{ m/s}$

## 7. Vectors in 2D — Variable Acceleration

For a particle moving in 2D, position is a vector function of time:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$$

Differentiate component-wise:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} \quad \mathbf{a} = \frac{d\mathbf{v}}{dt} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}$$

Speed (magnitude of velocity):  $|\mathbf{v}| = \sqrt{v_x^2 + v_y^2}$

**Example:** A particle has position  $\mathbf{r} = (t^3 - 3t)\mathbf{i} + (2t^2 - 1)\mathbf{j} \text{ m}$ . Find the speed at  $t = 2$ .

$$\mathbf{v} = (3t^2 - 3)\mathbf{i} + 4t\mathbf{j}$$

$$\text{At } t = 2: \mathbf{v} = (12 - 3)\mathbf{i} + 8\mathbf{j} = 9\mathbf{i} + 8\mathbf{j}$$

$$\text{Speed} = \sqrt{81 + 64} = \sqrt{145} \approx 12.0 \text{ m/s}$$

## 8. Key Formulae Summary

| Operation                  | Formula                               |
|----------------------------|---------------------------------------|
| Velocity from displacement | $v = \frac{ds}{dt}$                   |
| Acceleration from velocity | $a = \frac{dv}{dt}$                   |
| Alternative acceleration   | $a = v \frac{dv}{ds}$                 |
| Displacement from velocity | $s = \int v \, dt$ (use limits or +C) |
| Velocity from acceleration | $v = \int a \, dt$ (use limits or +C) |
| At rest                    | $v = 0$                               |
| Maximum speed              | $a = \frac{dv}{dt} = 0$               |
| 2D speed                   | $ \mathbf{v}  = \sqrt{v_x^2 + v_y^2}$ |

### Common Errors:

- Confusing displacement and distance — check for sign changes in  $v$ .
- Forgetting the constant of integration when integrating indefinitely — always apply an initial condition.
- Differentiating to find displacement (wrong direction) — differentiate for velocity/acceleration, integrate for velocity/displacement.
- In 2D, treat  $\mathbf{i}$  and  $\mathbf{j}$  components separately — never mix them.